

ADENIYI ADEYEMI SAMUEL

Back-End Focused Full Stack Engineer



[Github](#)



[Portfolio](#)



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Professional Summary

Highly skilled Software Engineer with over seven years of hands-on experience building, scaling, and deploying production-grade web, mobile, and AI-powered systems. Specialized in backend engineering using Node.js, PHP and Python (FastAPI), cloud-native architectures on AWS, and real-time applications. Proven track record of leading teams, optimizing performance, and delivering mission-critical systems used by over **30,000 users**.

Technical Skills & Engineering Depth

Programming Languages: JavaScript (ES6+), TypeScript, Python, SQL, PHP

Backend Engineering: Node.js, Express.js, NestJS, FastAPI, Laravel; RESTful API design, microservices architecture, authentication/authorization (JWT, OAuth2), background jobs, rate limiting, and API security hardening.

Frontend Engineering: React.js, Next.js, React Native; responsive UI design, component-driven architecture, state management, accessibility, and performance optimization.

AI & Intelligent Systems: Gemini API integration, AI inference pipelines, prompt engineering, ML model consumption, real-time AI-assisted workflows, automation systems.

Databases & Storage: PostgreSQL, MySQL, MongoDB, SupabaseDB, Redis; schema design, indexing, query optimization, migrations, data integrity, caching strategies.

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, ECS, CloudFormation), Docker, CI/CD pipelines (GitHub Actions), monitoring, logging, backups, and disaster recovery.

Engineering Practices: Clean Architecture, Domain-Driven Design principles, Agile/Scrum, technical documentation, code reviews, test-driven development, AI-assisted development.

Professional Work Experience

Waliin Social Networking Company

2025 – till date | Remote (US)

Senior Software Engineer (Backend-Focused)

Technology Stack: Nodejs, Next.js, Python (FastAPI), Laravel, PHP, Blade, HTML, CSS, Bootstrap, Git, Javascript.

Core Responsibilities

Backend Development (Primary Focus)

- Design, develop, and maintain scalable RESTful APIs using Node.js for user-facing features
- Build and optimize the API layer handling user data, content delivery, and real-time interactions
- Implement secure authentication and authorization systems (JWT/OAuth)
- Develop database schemas and optimize queries for MongoDB/PostgreSQL/MySQL
- Create real-time features using WebSockets/[Socket.io](#) for notifications and messaging
- Implement caching strategies with Redis/Memcached for performance optimization
- Ensure API security, data protection, and compliance with privacy regulations

Admin Panel Development

- Build and maintain the administrative interface using PHP CodeIgniter framework
- Develop admin features for content moderation, user management, and analytics
- Create dashboard components for monitoring platform metrics and KPIs
- Implement role-based access control for different admin permission levels
- Develop reporting tools and data visualization for admin insights.

Lead Full Stack Developer

2023 - 2024

University of Africa, Toru-Orua.

Lead development of enterprise-grade Edutech App/Portal that is widely used by over 10,000 students and staff

Key Achievements

- Led development of enterprise academic platform serving **10,000+ students & staff**
- Designed microservices architecture handling **1M+ daily requests**
- Improved system performance by **70%** through caching and query optimization
- Implemented CI/CD pipelines reducing deployment time by **80%**
- Mentored and led a team of 4 developers.

Full Stack Engineer (Contract) - CGI Nederland

2022 – 2023

- Developed and documented REST APIs with strong data modeling practices
- Built responsive frontend interfaces using React and Tailwind CSS
- Collaborated with cross-functional teams in Agile environment

Medical Technology

2020 - 2022

Front End Engineer

- Built responsive, accessible healthcare interfaces using React and Tailwind
- Created reusable UI component systems with Shadcn/UI
- Improved cross-browser consistency and UX reliability

Software Engineer I

TOKU-E Company · Full-time

2019 - 2020

Hybrid

Project for predicting changes in the Forex Market. I was responsible for the whole project and had the roles from project manager to architect the application and assumed Lead developer role. The application was distributed and scalable multi-components system.

1. Connecting via REST API's to external systems.
2. Hiring and managing other developers.
3. Designing the architecture and setting up the backend user's layer (mobile app) using nodejs and admin layer in PHP.

Skills: *PHP · MySQL · Spring Boot · Node.js · React.js · User Experience (UX) · MERN Stack*

AI-Enabled Projects (Production Grade)

AI-Powered Emergency Care EMS

[Live Demo](#)

[Github](#)

Tech Stack: TypeScript, Supabase, Node/Express.js, React.js, GeminiAI API, SQL, and others

- Developed an intuitive UI with better user's experience.
- Integration of GeminiAI API to power emergency first aid assistance.
- Supabase Auth and backend logic to handle server side/api requests.
- Created standard emergency report generation modules.

NYSC Toru-Orua Automated Portal

[Live Demo](#)

[Github](#)

Tech Stack: Next.js, Gemini API, SupabaseDB, SQL.

- Design and develop a robust NYSC-Toru-Orua portal to save, post community development services activities.

- AI-powered platform that suggest community development program that specifically address the unique needs of people in Toru-Orua community.
- Automation of all NYSC Toru-Orua Members activities increasing work efficiency by 80%.
- Tracking of previous administration contributions leading to safe record keeping and others.

Personal Projects

Geriatrics Vital Signs Monitor (Mobile Application)

[Live Demo](#)

[Github](#)

Tech Stack: React Native, Node/Express.js, SQL, SupabaseDB.

- Developed custom simulation engine with configurable parameters
- Used React Native's native modules for optimal performance
- Implemented local storage with automatic sync capabilities
- Use NodeJs to handle API requests from the arduino kits to the mobile app

Portfolio

[Live Demo](#)

[Github](#)

AWS & DevOps Experience

- Designed and maintained cloud architecture on AWS, utilizing EC2, S3, RDS, Lambda, and CloudFormation.
- Implemented monitoring with CloudWatch and automated backups for disaster recovery.
- Set up CI/CD pipelines using GitHub Actions and AWS CodeDeploy for seamless deployments.
- Managed containerized applications with Docker and AWS ECS.

AI-Assisted Development

- Regularly use GitHub Copilot, Claude Code, Cursor, and Windsurf to accelerate development, reduce boilerplate code, and improve code quality.
- Implemented AI-assisted code review and documentation automation in team workflows.

Languages and Technologies

- **Languages:** JavaScript, TypeScript, Python, SQL, PHP
- **Frameworks & Tools:** React, Node.js, AWS, Azure Cloud, Docker, Git, Supabase, MongoDB, FastAPI, TensorFlow, Scikit-learn
- **Soft Skills:** Leadership, Project Management, Agile Development, Technical Documentation

Education, Practices & Professional Development

Bachelor of Science in Computer Science — University of Ilorin, Nigeria (2024)

Relevant Coursework: Data Structures, Algorithms, Artificial Intelligence, Machine Learning, Software Engineering.

Engineering Practices: Clean Architecture, Agile/Scrum, technical documentation, mentoring, AI-assisted development using GitHub Copilot, Cursor, Claude, and Windsurf.

Professional Strengths: System design, backend scalability, cross-team collaboration, technical leadership, and ownership of production systems.